Multi-Scale Weather Forecasting Using Deep Learning Architectures With Chennai Climate Data

Abstract

Weather forecasting is important for climate-sensitive activities such as agriculture, water-resource management, disaster risk management, energy planning, and urban planning. Short-range forecasts support immediate decisions, while medium- and long-range forecasts support strategic planning. Traditional forecasting approaches may have difficulty capturing the nonlinear, nonstationary, multivariate, and scale-dependent characteristics of weather data. This paper investigates a deep learning framework for multi-scale weather forecasting using Chennai-specific historical weather data. The study compares three architectures: Long Short-Term Memory (LSTM), LSTM-CNN, and LSTM-Transformer, for predicting temperature, humidity, and rainfall across daily, weekly, and monthly forecasting scales. Hourly weather data for Chennai from 2000 to 2025 were obtained through the Open-Meteo API and resampled into daily, weekly, and monthly datasets. The data were normalized and prepared using a sliding-window sequence approach for multi-step and multivariate prediction. A walk-forward validation strategy was used to evaluate model performance, while Keras Tuner was applied for automated hyperparameter optimization. The LSTM model captures temporal dependencies, the CNN component helps identify local temporal patterns, and the Transformer component uses attention mechanisms to capture longer-range dependencies. Experimental results show that LSTM-CNN provides strong performance for short-term forecasting, particularly for temperature and humidity, while LSTM-Transformer is more suitable for longer forecasting horizons because of its ability to capture global temporal dependencies. Rainfall remains the most difficult variable to forecast because of its intermittent, bursty, and highly nonlinear behavior. The study demonstrates that no single architecture is uniformly optimal across all temporal scales and that model selection should depend on the forecasting horizon and characteristics of the target variable. The proposed multi-scale framework provides a scalable and region-specific approach for weather prediction and can support practical applications such as agricultural scheduling, water-resource planning, disaster preparedness, and climate-resilient decision-making.

Keywords

Deep learning, Keras Tuner optimization, LSTM neural networks, multi-scale weather forecasting, time series prediction